1) Reescrevendo a EDO, com séries :

$$\sum_{n=2}^{\infty} n(n+1) c_n x^{n-2} - (1+x)\sum_{n=0}^{\infty} c_n x^{n} = 0 \rightarrow$$

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0 \rightarrow$$

$k = n-2 \qquad k = n \qquad k = n+1$

reescrevendo em função de K:

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^{k} - \sum_{k=0}^{\infty} c_k x^{k} - \sum_{k=1}^{\infty} c_{k-1} x^{k} = 0 \rightarrow$$

removendo dos termos (um de cada $k=0$):

$$2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^{k} - c_0 - \sum_{k=1}^{\infty} c_k x^{k} - \sum_{k=1}^{\infty} c_{k-1} x^{k} = 0 \rightarrow$$

$$2c_2 - c_0 + \left(\sum_{k=1}^{\infty} \left((k+2)(k+1) c_{k+2} - c_k - c_{k-1} \right) x^{k} \right) = 0$$

$\therefore$ $c_2 = \frac{c_0}{2}$, e a recorrência fica: $c_{k+2} = \dfrac{c_{k+1} + c_k}{(k+2)(k+1)}$

$\forall\, k \geq 1$. Calculando alguns coeficientes: $\left(c_2 = \frac{c_0}{2} \right)$

Para k:

k		
1	c_3	$\frac{c_1 + c_0}{3 \cdot 2}$
2	c_4	$\frac{c_2 + c_1}{4 \cdot 3} = \frac{c_0 + 2c_1}{4 \cdot 3 \cdot 2}$
3	c_5	$\frac{c_3 + c_2}{5 \cdot 4} = \frac{\frac{c_1 + c_0}{3 \cdot 2} + \frac{c_0}{2}}{5 \cdot 4} = \frac{c_1 + 4c_0}{5 \cdot 4 \cdot 6}$
4	c_6	$\frac{c_4 + c_3}{6 \cdot 5} = \frac{\frac{c_0 + 2c_1}{4 \cdot 3 \cdot 2} + \frac{c_1 + c_0}{3 \cdot 2}}{6 \cdot 5} = \frac{6c_1 + 5c_0}{6 \cdot 5 \cdot 4 \cdot 6}$
⋮	⋮	⋮

Logo a solução geral é da forma:

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 \dots$$, substituindo:

$$y = c_0 + c_1 x + \frac{c_0}{2} x^2 + \frac{c_0 + c_1}{6} x^3 + \frac{c_0 + 2c_1}{24} x^4 + \frac{4c_0 + c_1}{120} x^5 \dots$$

agrupando:

$$y = c_0 \underbrace{\left(1 + \frac{1}{2} x^2 + \frac{1}{6} x^3 + \frac{1}{24} x^4 + \frac{1}{30} x^5 \dots \right)}_{y_0} + c_1 \underbrace{\left(x + \frac{1}{6} x^3 + \frac{1}{12} x^4 + \frac{1}{120} x^5 \dots \right)}_{y_1}$$

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